EpiLink: an interpretable pairwise compatibility model for recent transmission linkage and clustering

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Abstract

Genomic epidemiology is widely used in outbreak investigation, but current approaches often force a trade-off between detailed transmission reconstruction and simple threshold-based clustering. Bayesian transmission-tree methods can be highly informative yet are often too computationally demanding for routine real-time surveillance, while threshold-based approaches rely on context-specific cut-offs that can limit robustness and interpretability. We developed EpiLink, a pairwise linkage model that integrates temporal and genetic data into an interpretable compatibility score for recent transmission, defined as either direct transmission or co-primary infection from a common source. We evaluated EpiLink on a large synthetic SARS-CoV-2 transmission tree, examining pairwise discrimination, recovery of local transmission structure through graph community detection, and temporal stability of inferred clusters. We benchmarked EpiLink against a logistic regression model trained on labelled transmission pairs, and assessed robustness under systematic perturbations to key epidemiological and evolutionary parameters under both matched and misspecified inference conditions. Under baseline conditions, the logistic model achieved the strongest pairwise discrimination, while the best-performing EpiLink variant reached comparable cluster recovery. Graph sparsification retained nearly all total edge weight while discarding the large majority of edges, yielding substantial runtime reductions with no meaningful effect on downstream community structure. Cluster partitions were temporally stable across consecutive weekly transitions, with stability rising rapidly once case accumulation exceeded the earliest sparse weeks. In the Boston dataset, EpiLink recovered clusters corresponding to the documented conference and skilled nursing facility outbreaks without requiring labelled training data. Together, these results suggest that EpiLink provides a practical and interpretable bridge between computationally intensive transmission reconstruction and simpler threshold-based clustering. It is most useful when consensus genomes and sampling times are available, labelled training data are absent, and the goal is to identify recent transmission clusters. EpiLink is available at `https://github.com/ydnkka/epilink`.

Author summary

When investigating an outbreak, public health officials need to identify which patients are likely connected—either because one infected the other, or because both caught the illness from the same source. Current tools often use fixed rules, such as requiring cases to occur within the same week or for their pathogen genomes to differ by fewer than a

certain number of changes. These cut-offs are fast and simple, but they can miss real connections or flag spurious ones, and different teams apply different rules, making comparisons across settings difficult. We developed EpiLink, a method that replaces fixed thresholds with a biologically grounded score. For each pair of cases, EpiLink asks: are the timing gap and the degree of genomic similarity jointly consistent with what we would expect if these two people were recently linked by transmission? The score is transparent—it reflects explicit assumptions about how infections progress and how genomes accumulate changes over time, and those assumptions can be inspected and adjusted. We tested EpiLink on simulated COVID-19 outbreaks and on a real dataset from Boston. It successfully recovered groups of patients linked to documented superspreading events, such as a large conference and a skilled nursing facility, without needing any labelled examples to train on. Cluster assignments were also stable as new cases accumulated over time, which is important for real-time surveillance. EpiLink is most useful when patient genomes and sample dates are available, labelled training data are unavailable, and the goal is to identify recent transmission clusters in an interpretable and reproducible way.

Introduction

Identifying clusters of epidemiologically linked infections is a core task in infectious disease surveillance, underpinning outbreak investigation, situational awareness, and targeted public health intervention. Increasingly, this task is supported by pathogen genomic data, which can provide high-resolution information on relatedness between infections and has become an important component of outbreak detection and transmission analysis in public health surveillance [1, 2]. This task is particularly challenging in outbreaks shaped by superspreading events, in which a small number of infections give rise to a disproportionately large number of secondary cases [3]. In such settings, rapid bursts of transmission, incomplete sampling, and limited pathogen genetic diversity can make it difficult to distinguish true transmission links from coincidental similarity, especially when relying on sequence data alone [4, 5]. Superspreading has been documented across a range of pathogens, including SARS-CoV-1, MERS, Ebola, and SARS-CoV-2 [6–10], and is now recognised as a fundamental driver of transmission heterogeneity. Despite its epidemiological importance, the timely identification and characterisation of superspreading-driven molecular transmission clusters remain methodologically unresolved, particularly in the context of large-scale genomic surveillance.

Current analytical approaches often force a trade-off between detailed transmission reconstruction and simple threshold-based clustering. In many surveillance settings, however, the relevant question is narrower than full transmission-tree inference: given the observed genomes and sampling times for two cases, are they plausibly linked by recent transmission? This question is particularly salient in superspreading-driven outbreaks, where many cases may accumulate within a short time window and share minimal genetic divergence. Methods such as outbreaker2, TransPhylo, and SCOTTI can explicitly model unsampled cases and uncertainty, yet they are generally best suited to detailed outbreak reconstruction and require assumptions and computational investment that may be difficult to support in real-time public health settings [11–15]. By contrast, fixed-threshold and distance-based clustering approaches remain attractive because they are fast, scalable, and easy to implement [16–21].

Their practical appeal, however, comes at the cost of interpretability. Small genetic distances do not map cleanly onto recent transmission, particularly when pathogen diversity is low, mutation processes are stochastic, sampling is incomplete, or sampling times carry information that static genetic cut-offs ignore [4, 5, 17, 18, 21]. As a result,

the epidemiological meaning of a pairwise link, or of a cluster derived from many such links, is often unclear. This limitation also applies to more flexible network-based strategies. Although hierarchical thresholding and community detection can reduce some of the instability associated with single cut-offs [22, 23], they still depend on underlying pairwise similarities whose relationship to recent transmission may be weak or ambiguous. The central challenge, therefore, is not only to detect structure in genomic data, but to do so in a way that yields clusters with a clear and defensible epidemiological interpretation.

A key part of this challenge is how pairwise genetic distances should be interpreted in biological and epidemiological terms when transmission is recent and highly clustered. Two sampled genomes separated by only a few single-nucleotide polymorphisms may suggest close relatedness, but they do not by themselves identify the exact relationship. Instead, that similarity may reflect direct infection, transmission through one or more unsampled intermediates, or infection from a common source followed by limited divergence across separate chains [24]. This ambiguity is especially pronounced for SARS-CoV-2 and other pathogens characterised by short generation intervals, rapid epidemic expansion, and relatively slow short-term sequence evolution [5, 18, 24–26]. In such settings, consensus genomes often remain identical or nearly identical across many epidemiologically distinct cases, so that genetic similarity alone provides only weak evidence about recent transmission at the scale relevant to superspreading events. While the incorporation of temporal data can improve resolution, current methods often rely on fixed cut-offs [27] or supervised models trained on labelled transmission pairs [20], and therefore lack a principled way to carry uncertainty in timing, infectiousness, and mutation processes through to cluster inference.

To address this interpretability gap, we developed EpiLink, a threshold-free, process-based method for evaluating recent-transmission compatibility between sampled cases. EpiLink does not seek to infer a fully resolved transmission tree. Instead, it assesses whether the observed temporal separation and genetic distance between two cases are consistent with Monte Carlo draws generated under a user-defined set of latent recent-transmission scenarios. This formulation is designed for the operational scale at which surveillance often functions, where the aim is not necessarily to identify an exact infector, but to determine whether two cases plausibly belong to the same recent transmission neighbourhood. By explicitly modelling uncertainty in infection timing, infectiousness, unsampled transmission, and mutation accumulation, EpiLink produces pairwise compatibility scores with a clear mechanistic interpretation, while avoiding reliance on arbitrary distance thresholds or labelled training data.

In this study, we evaluate EpiLink using both synthetic outbreak data and a published SARS-CoV-2 dataset from Boston [28]. We first examine the recent-transmission compatibility surface induced by the model across combinations of temporal and genetic distance, and assess the extent to which graph sparsification can reduce network density while preserving informative pairwise structure. We then evaluate pairwise discrimination and graph-based cluster recovery under baseline conditions and systematic perturbations of key simulation parameters, and assess the stability of inferred clusters as surveillance data accumulate. Finally, we test whether the method recovers epidemiologically documented superspreading structure in the Boston dataset. Throughout, we compare EpiLink with logistic regression trained on labelled pairs as a supervised reference.

Materials and methods

EpiLink compatibility model

We developed EpiLink, a pairwise compatibility model that assesses whether the observed consensus genetic distance g_{ij} and sampling-time difference t_{ij} between two sampled cases (i, j) are consistent with one or more candidate latent transmission histories (Fig 1). Rather than inferring a unique transmission tree, EpiLink evaluates compatibility with hypothesised epidemiological relationships while accounting for uncertainty in infection timing, testing delay, and mutation accumulation.

We considered two classes of latent transmission scenarios up to a maximum hidden depth M:

$$\mathcal{S}_M = \big\{H_{\mathrm{AD}}(m) : m = 0, \ldots, M\big\} \cup \big\{H_{\mathrm{CA}}(m_i, m_j) : m_i, m_j \geq 0,\ m_i + m_j \leq M\big\}.$$

Here, $H_{\mathrm{AD}}(m)$ denotes an ancestor-descendant scenario in which case j descends from case i through m unsampled intermediates, and $H_{\mathrm{CA}}(m_i, m_j)$ denotes a common-ancestor scenario in which both cases descend independently from a shared unsampled source. The cases $H_{\mathrm{AD}}(0)$ and $H_{\mathrm{CA}}(0, 0)$ correspond to direct transmission and co-primary infection, respectively. In the default configuration used throughout this study, the target subset is $\mathcal{S}_\star = \{H_{\mathrm{AD}}(0), H_{\mathrm{CA}}(0, 0)\}$, implying a maximum depth of $M = 0$.

For each scenario $s \in \mathcal{S}_M$, EpiLink generates Monte Carlo draws from a shared natural-history model with Gamma-distributed latent (E), presymptomatic infectious (P), and symptomatic infectious (I) stages, together with a Gamma-distributed delay from symptom onset to testing [29–31]. Writing the incubation period as $\tau_{\mathrm{inc}} = y_E + y_P$ and the generation time as $\tau_{\mathrm{gen}} = \tau_{\mathrm{inc}} + y_{\mathrm{toit}}$ (where y_{toit} is the time from onset of infectiousness to transmission), these draws induce a scenario-specific distribution for the expected testing-time difference T_s. The same latent history defines an effective transmission-related branch length B_s: under ancestor-descendant scenarios, this is the sum of generation intervals along the chain; under common-ancestor scenarios, it is the sum of lineage lengths from the shared source to both sampled descendants. The full derivation of T_s and B_s is given in S1 Appendix.

Branch-length draws were converted into expected genetic distances using a per-genome daily substitution rate $\lambda = \nu L/365$, where ν is the substitution rate per site per year and L is the genome length. Under a deterministic clock, $G_s = \lambda B_s$; under a stochastic (Poisson) mutation process, $G_s \mid B_s \sim \mathrm{Poisson}(\lambda B_s)$. A relaxed uncorrelated log-normal clock draws r from a log-normal distribution [32].

Compatibility was quantified by locating each observed value within the corresponding scenario-specific empirical distribution. For observed value x_{obs} and n_s Monte Carlo draws from scenario s:

$$p_s(x_{\mathrm{obs}}) = \frac{\#\{i : x_i^{(s)} \leq x_{\mathrm{obs}}\}}{n_s}, \qquad C_s(x_{\mathrm{obs}}) = 1 - 2\,|p_s(x_{\mathrm{obs}}) - 0.5|\,.$$

Temporal and genetic compatibilities $C_{T,s}(i, j)$ and $C_{G,s}(i, j)$ are combined as their product to obtain the scenario-level score $C_s(i, j) = C_{T,s}(i, j)\, C_{G,s}(i, j)$, and then summed across the target subset:

$$C_{\mathcal{S}_\star}(i, j) = \sum_{s \in \mathcal{S}_\star} C_s(i, j).$$

This is a compatibility model rather than a calibrated probabilistic transmission model: scores reflect how centrally the observed pair sits within the simulated distributions induced by $\mathcal{S}_\star$, not posterior transmission probabilities.

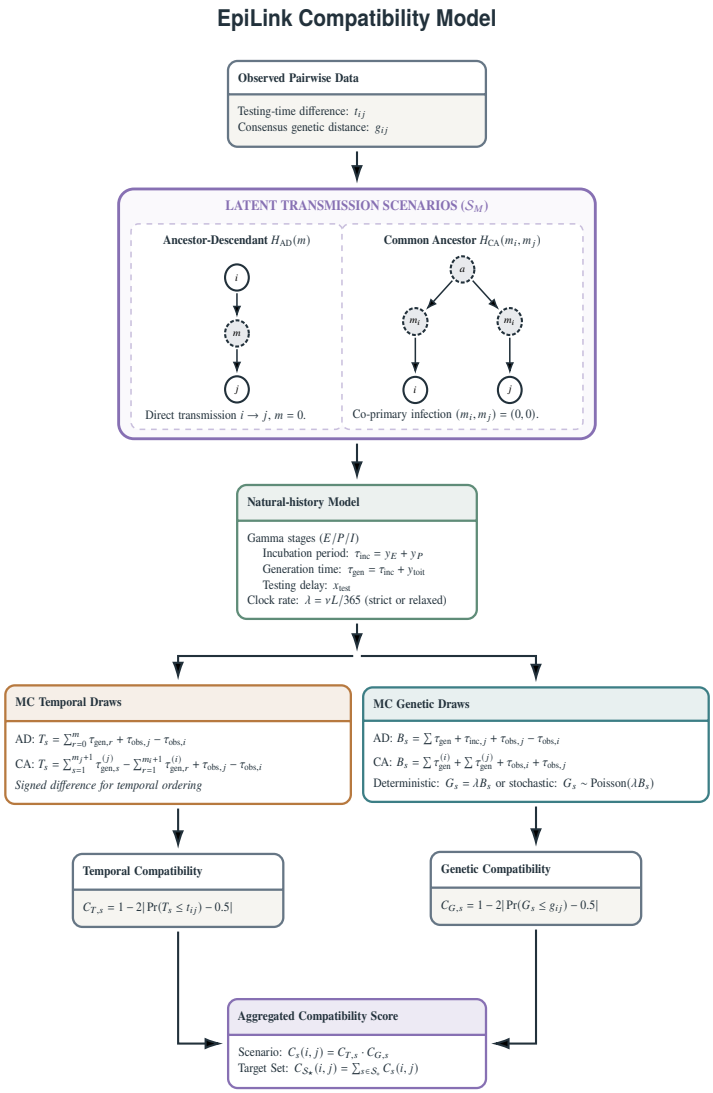

Fig 1. Overview of the EpiLink compatibility model. For a sampled pair (i, j), the observed inputs are the testing-time difference t_{ij} and the consensus genetic distance g_{ij}. These are evaluated against a set of latent transmission scenarios $\mathcal{S}_M$, encompassing ancestor-descendant chains $H_{\mathrm{AD}}(m)$ and common-ancestor histories $H_{\mathrm{CA}}(m_i, m_j)$. A shared natural-history model with Gamma-distributed latent (y_E), presymptomatic infectious (y_P), and symptomatic infectious (y_I) stage durations, together with a Gamma-distributed delay from symptom onset to testing (x_{test}) drives two parallel Monte Carlo tracks: the temporal track samples the implied testing-time difference T_s for each scenario, and the genetic track samples expected substitution counts G_s under a deterministic ($G_s = \lambda B_s$) or Poisson ($G_s \sim \mathrm{Poisson}(\lambda B_s)$) mutational process. y_{toit} is the time from onset of infectiousness to transmission; ν is the substitution rate per site per year; L is the genome length; B_s is the implied effective transmission-related branch length; and $\tau_{\mathrm{obs}} = \tau_{\mathrm{inc}} + x_{\mathrm{test}}$ is the total duration from exposure to testing. Temporal compatibility $C_{T,s}$ and genetic compatibility $C_{G,s}$ are each computed as two-sided percentile proximity scores. The final target compatibility $C_{\mathcal{S}_\star}(i, j)$ is the product summed over the user-defined target scenario subset $\mathcal{S}_\star$.

Synthetic benchmark dataset

We evaluated EpiLink using synthetic outbreak datasets derived from SCoVMod, a large-scale individual-based SARS-CoV-2 transmission simulator [33]. Starting from SCoVMod infection-history and transmission-event outputs, we constructed a directed transmission network, resolved multiply infected nodes by retaining the earliest recorded parent, and selected a weakly connected component containing approximately 5,000 nodes. A rooted arborescence was then obtained by computing the maximum spanning arborescence with edge weights defined as the negative simulation timestep, thereby favouring earlier parental assignments. The resulting transmission tree comprised 4,990 nodes and 4,989 directed edges.

Transmission heterogeneity in the underlying SCoVMod outbreak was summarised by fitting a negative-binomial offspring distribution to the full transmission network, yielding an estimated dispersion parameter of $\hat{k} = 0.233$ (95% CI: 0.232–0.235) and mean reproductive number $\bar{R}_t \approx 1.00$ (95% CI: 0.992–1.006). Approximately 15.6% of individuals accounted for 80% of onward transmission. Synthetic epidemic dates and pairwise genetic distances were generated by overlaying the EpiLink natural-history model onto this fixed transmission-tree topology.

Baseline natural-history parameters

Key baseline evaluation parameters (Table 1) were drawn from published SARS-CoV-2 estimates [29, 34, 35]. Other $E/P/I$ infectiousness profile parameters were kept fixed at their fitted values [29]. To aid interpretation, the mean and coefficient of variation (CV) of the Gamma distributions are shown. In implementation, these are converted into the corresponding shape k and scale θ parameters. Multipliers of $\{0.75, 1.25\}$ were applied to strictly positive parameters (incubation mean and CV, testing-delay mean and CV, and substitution rate), whereas multipliers of $\{0.0, 1.25\}$ were applied to the clock-relaxation parameter (log-normal std), allowing inclusion of a strict-clock case. In total, this produced 12 perturbation scenarios.

Table 1. Baseline evaluation parameters and perturbation design.

Parameter	Baseline value	Lower mult.	Upper mult.
Incubation mean	5.505 days	0.75	1.25
Incubation CV	0.415	0.75	1.25
Testing-delay mean	1.0 day	0.75	1.25
Testing-delay CV	1.0	0.75	1.25
Substitution rate	10^{-3} sub/site/year	0.75	1.25
Clock relaxation	0.33	0.0	1.25
Fixed simulation parameters:			
Synthetic genome length	5,000 bp	—	—
Monte Carlo draws	10,000	—	—
Target subset	$\{H_{\mathrm{AD}}(0), H_{\mathrm{CA}}(0,0)\}$	—	—
Sparsification threshold	10^{-4}	—	—

Model variants

Synthetic pairwise temporal and genetic data—either deterministic (D) or stochastic (S)—were supplied as input to EpiLink, which itself can apply either a deterministic or stochastic formulation for compatibility score inference. This yields four EpiLink variants: EDD (deterministic inference, deterministic input), EDS (deterministic

inference, stochastic input), ESD (stochastic inference, deterministic input), and ESS (stochastic inference, stochastic input). Two logistic regression models (LD and LS) were fitted directly to the corresponding pairwise temporal and genetic features; both were trained on a random 10% subsample of all case pairs and are treated throughout as supervised reference comparators.

Simulation design and sensitivity analysis

We conducted a simulation study to evaluate inference performance and robustness under a range of epidemiological and evolutionary conditions. Synthetic outbreak datasets were generated on the fixed 4,990-case transmission tree under the baseline parameterisation described above. Sensitivity analyses were designed to assess the effect of perturbing the two core mechanisms governing pairwise compatibility:

1. **Infection-to-sampling interval**, varied through the mean and coefficient of variation of the incubation-period and testing-delay distributions;
2. **Molecular clock rate**, varied through the median substitution rate and degree of relaxation.

Perturbations were applied one at a time, and each scenario was evaluated under two inference conditions to distinguish performance changes due to altered data-generating conditions from those due to model misspecification.

In the **matched** condition, inference parameters were updated to match the perturbed data-generating values. This represents best-case performance under each altered regime, assuming perfect knowledge of the underlying epidemiological and molecular parameters.

In the **mismatched** condition (*generation-varied, inference-fixed*), data-generation parameters were perturbed as above, but inference parameters were held fixed at their baseline values. This condition quantifies the robustness of the inference procedure to parameter misspecification.

The logistic regression comparators were treated analogously. Under the matched condition, they were retrained separately for each perturbation scenario. Under the mismatched condition, they were trained once using baseline data only and then applied unchanged to perturbed scenarios, thereby assessing performance under distribution shift.

Performance losses were calculated as relative changes from the unperturbed matched baseline; comparing matched and mismatched results separates the effect of the altered generating regime from the additional cost of inference under incorrect assumptions.

Sparsification analysis

We identified minimum score thresholds at which nearly all total edge weight was retained while graph density was substantially reduced. For each model, the optimal sparsification threshold was defined as the smallest value in $\{0, 10^{-4}, 10^{-3}, 10^{-2}, 10^{-1}\}$ that retained at least 99.5% of total edge weight.

Graph clustering and evaluation metrics

For all synthetic experiments, weighted graphs were constructed from pairwise compatibility scores (or logistic probabilities) after excluding edges below the chosen sparsification threshold (10^{-4}). Community detection was then performed using the Leiden algorithm [36] with a modularity-based objective. The resolution parameter was

swept over $\gamma \in \{0.1, 0.2, \ldots, 1.0\}$, with 10 random restarts at each value. For each model, the resolution giving the highest BCubed F1 score was selected.

Clustering performance was assessed using BCubed precision and recall [37] against overlapping reference neighbourhoods derived from the transmission tree. For each node u, the reference cluster consisted of u together with all of its direct infectees. BCubed precision measures, for each case, the proportion of inferred co-cluster members that truly share its reference neighbourhood, whereas BCubed recall measures the proportion of the reference neighbourhood recovered in the inferred cluster. BCubed F1 score is the harmonic mean of precision and recall, averaged across all cases.

Pairwise discrimination was quantified using the average precision (AP) because positive pairs were rare, comprising approximately 0.14% of all sampled pairs, making precision–recall summaries more informative in this highly imbalanced setting [38]. A pair was labelled positive if they were connected by direct transmission or were sibling infectees sharing the same parent in the transmission tree.

Temporal stability analysis

Temporal stability was evaluated by grouping sampled cases into weekly availability bins. After each cumulative weekly update, clustering was recomputed using only the cases available up to that time point. Consecutive weekly partitions were then compared using three measures: forward overlap, defined as the fraction of a later cluster recovered from its best-matched earlier cluster; backward overlap, defined as the fraction of an earlier cluster retained in its best-matched later cluster; and the Jaccard index, which provides a symmetric measure of stability that penalises both splitting and merging. For the logistic regression reference models, training labels were restricted to the first three cumulative weekly bins (86 initial cases), reflecting a limited early-supervision setting. The clustering resolution for all models was fixed using the early window and then held constant across subsequent cumulative analyses.

Boston empirical dataset and clustering

The empirical evaluation used 772 high-quality SARS-CoV-2 whole-genome sequences (greater than 98% coverage) collected by Massachusetts General Hospital (MGH) and the Massachusetts Department of Public Health (DPH) during the first wave of the COVID-19 pandemic in Boston, Massachusetts (March–June 2020) [28]. Sequences were aligned using Nextclade against the SARS-CoV-2 reference genome (GenBank accession no. MN908947.3) [39]. Pairwise consensus SNP distances were computed under the TN93 substitution model [40]. Sequence metadata included collection date and an exposure category label (Conference, SNF, BHCHP, City, Other) derived from epidemiological investigation. EpiLink scores were computed for all 297,396 unique pairs using the stochastic variant (ES) under baseline inference parameters. A weighted graph was constructed with minimum edge weight 10^{-4}, and the Leiden algorithm was run at resolution $\gamma = 0.3$ with 10 restarts. Clusters of size ≥ 2 were retained. Focus clusters were those containing at least one Conference- or SNF-linked case; deviation from background exposure prevalence was tested using a χ^2 goodness-of-fit test.

Software and data availability

The EpiLink software is available at `https://github.com/ydnkka/epilink`. Code, configuration files, and derived outputs required to reproduce the synthetic and empirical analyses presented here are available at `https://github.com/ydnkka/epilink/evaluation`. The Boston empirical inputs

were derived from the previously published dataset [28], and the synthetic benchmark was based on SCoVMod outputs [33].

For journal submission, replace with the final journal-formatted Data Availability Statement, including the persistent repository link and archived release information.

Results

EpiLink compatibility surfaces and graph sparsification

Across the joint genetic and temporal space, compatibility scores were highest for pairs with small SNP distances and short sampling gaps. They declined steadily as separation increased in either dimension (Fig. 2). The surface showed two broad high-compatibility regimes corresponding to the two target scenarios: one with very short temporal gaps and modest genetic distance, consistent with co-primary infection, and a second region with slightly longer temporal gaps but minimal genetic distance, consistent with direct transmission. Under the stochastic formulation (Fig. 2B), these regions were broader and less sharply bounded than under the deterministic formulation (Fig. 2A), indicating greater tolerance to small deviations around the most compatible combinations. Overall, the results show that EpiLink assigns the greatest support to case pairs whose observed genetic and temporal distances are plausibly explained by either direct transmission or co-primary infection from a shared source, while strongly down-weighting pairs that fall outside these ranges.

The evaluation used a 4,990-node transmission tree yielding 12,447,555 scored pairs, the majority of which had negligible compatibility scores and contributed little to community structure while substantially increasing computational cost. We therefore examined the impact of removing low-weight edges.

Across all six models—EDD (deterministic inference, deterministic input), EDS (deterministic inference, stochastic input), ESD (stochastic inference, deterministic input), ESS (stochastic inference, stochastic input), LD (logistic regression, deterministic input), and LS (logistic regression, stochastic input)—we identified 10^{-4} as a reasonable sparsification threshold. At this threshold, only 2.9% to 13.1% of scored edges were retained, yet more than 99.8% of total graph weight was preserved. Thus, the 87–97% of uninformative edges accounted for less than 0.001% of total graph weight. This reduced pipeline runtime from approximately 38–62 seconds to 0.9–3.6 seconds per model, corresponding to a 17–54× speed-up, with no meaningful effect on downstream community structure. We therefore used this threshold in all subsequent clustering analyses.

Overall synthetic performance across all six models

In the baseline scenario, LD showed the strongest overall performance, achieving the highest average precision (AP = 0.568) and F1 score (0.688), while also maintaining high partition stability across Leiden resolution steps (mean = 0.895, SD = 0.044) (Table 2). LS produced lower discrimination than LD (AP = 0.275; F1 = 0.570), but had the highest mean partition stability overall (0.927), indicating that its inferred partitions were especially robust to the choice of clustering resolution. Among the EpiLink variants, ESD performed best, with AP = 0.469, F1 = 0.601, and mean partition stability = 0.851 (SD = 0.073). EDD and ESS showed intermediate performance, with F1 scores of 0.556 and 0.534, respectively. EDS had the weakest discrimination at baseline (AP = 0.090), despite still achieving a F1 score of 0.476 and a mean stability of 0.834. Overall, these baseline results show that weaker edge-level discrimination did not always translate proportionally into uninformative clustering.

Table 2. Performance at baseline scenario.

Model	Average precision (AP)	BCubed F1 score	Mean partition stability (SD)
EDD	0.300	0.556	0.871 (0.077)
EDS	0.090	0.476	0.834 (0.090)
ESD	0.469	0.601	0.851 (0.073)
ESS	0.190	0.534	0.830 (0.079)
LD	0.568	0.688	0.895 (0.044)
LS	0.275	0.570	0.927 (0.094)

The broader trend across all twelve perturbed scenarios was consistent with the baseline ordering (Fig. 3). LD remained the top-performing model in discrimination and clustering accuracy under both matched and mismatched conditions, while ESD was the strongest-performing EpiLink variant across scenarios. EDS consistently showed the lowest AP, yet still yielded moderately coherent partitions, reinforcing the point that aggregating many noisy pairwise scores into a graph can still recover meaningful local transmission neighbourhoods. Across all six models, matched and mismatched conditions followed closely similar trajectories for AP, F1, and partition stability, indicating that parameter misspecification had a limited effect on overall model performance.

Robustness to parameter misspecification

We evaluated the sensitivity of F1 performance to perturbations of data-generating parameters under both matched and mismatched inference conditions. Under the matched condition, F1 losses were generally smaller in magnitude than the corresponding AP losses (Fig 4, S1 Fig), with the majority of scenarios producing effects within ± 0.10 of baseline for all models. This indicates that all six models maintain reasonable clustering performance when inference is appropriately adapted to reflect shifts in the underlying epidemiological or evolutionary process.

The mismatched condition revealed substantially greater and more heterogeneous performance degradation. The most severe losses occurred under increased incubation CV, where all models suffered F1 reductions of 0.56–0.66, by far the largest effect observed across any scenario. This pattern was unique to the upward perturbation; reducing incubation CV produced minimal change under either condition, indicating an asymmetric sensitivity to this parameter. Reduced clock rate constituted the second major vulnerability in the mismatched condition, with F1 losses of 0.27, 0.15, and 0.13 for EDD, ESD, and EDS respectively, while the logistic regression models LD and LS were comparatively unaffected (losses of approximately 0.11 and 0.02), consistent with these models not explicitly incorporating a molecular clock assumption. Reduced incubation mean also produced notable mismatched degradation, particularly for EDD (-0.20) and EDS (-0.12), whereas increasing the incubation mean had little effect in either condition.

Perturbations to testing delay parameters produced consistently small F1 changes across all models and both conditions, suggesting that moderate misspecification of the testing delay distribution is relatively inconsequential for clustering performance. Similarly, the clock relaxation scenarios—applying strict clock or increasing the relaxation—had largely negligible effects, with the exception of a modest matched-condition loss for EDS under the strict clock scenario.

Comparing models, EDD showed the greatest overall sensitivity to mismatched perturbations, while the logistic regression models (LD and LS) exhibited the most consistently robust performance across scenarios. Stochastic input models (EDS, ESS,

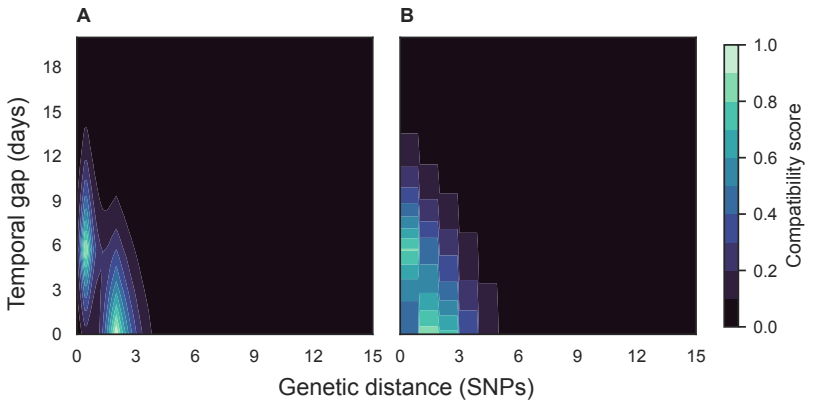

Fig 2. EpiLink compatibility surfaces. Contour plots of EpiLink compatibility as a function of genetic distance (x-axis) and temporal gap (y-axis), using baseline inference parameters. (A) deterministic genetic component; (B) stochastic genetic component.

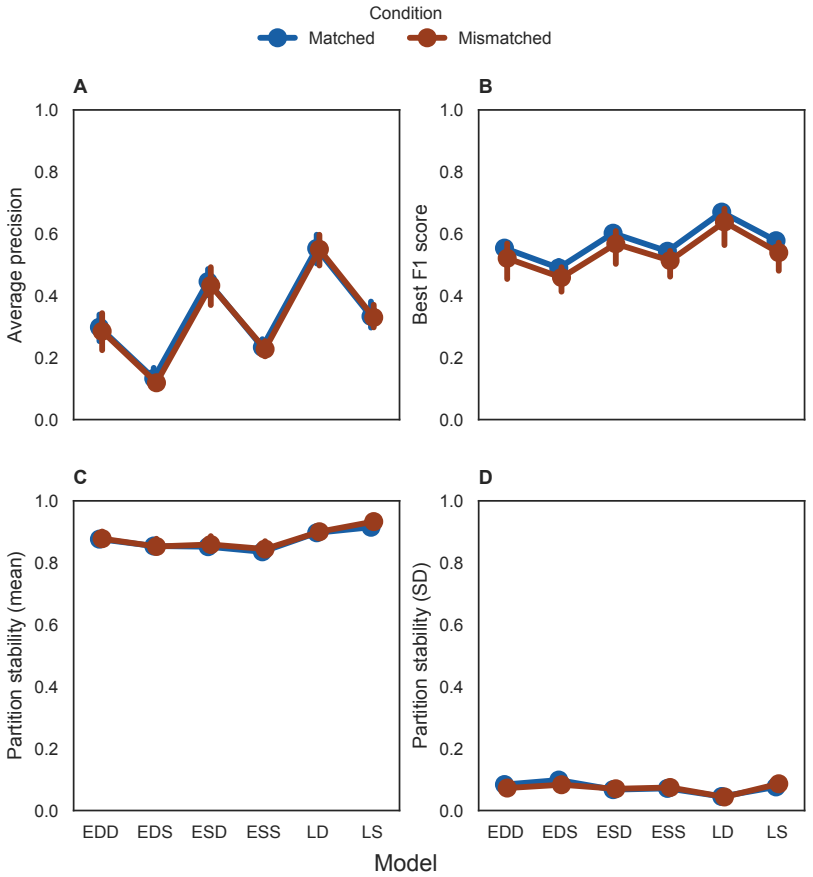

Fig 3. Overall performance across all six models. Performance metrics under matched (inference parameters updated to reflect each perturbed data-generation scenario) and mismatched (inference parameters held fixed at baseline; data-generation parameters varied) conditions, evaluated across all 12 perturbation scenarios plus the unperturbed baseline. (A) Average precision (AP). (B) Best F1 score across Leiden resolution sweep. (C) Mean resolution stability (F1 between consecutive Leiden resolution steps; measures robustness to the choice of resolution parameter). (D) SD of resolution stability. Error bars show the 95% confidence interval.

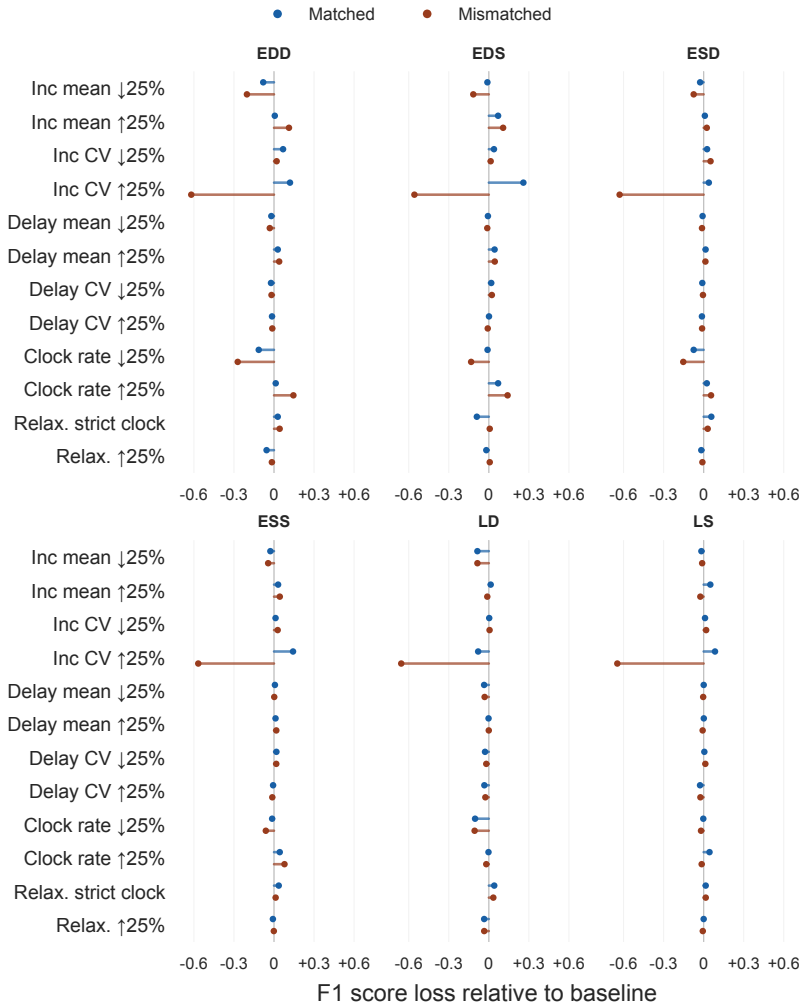

Fig 4. Sensitivity of F1 performance to perturbations of data-generating parameters. Lollipop plot showing best-F1 loss relative to the unperturbed baseline for each model across twelve perturbation scenarios, under matched (blue) and mismatched (coral) inference conditions. Matched: inference parameters updated to reflect each perturbed scenario; mismatched: inference parameters held fixed at baseline. Each row corresponds to a perturbation scenario; the multiplier applied to the indicated parameter is shown on the left. Positive values indicate improved performance relative to baseline; negative values indicate degradation.

LS) tended to show smaller degradations than their deterministic-input counterparts (EDD, ESD, LD) in several mismatched scenarios. Overall, these results indicate that model robustness to parameter misspecification is highly scenario-dependent, with incubation period variability and molecular clock rate representing the primary sources of performance vulnerability.

Temporal stability of cumulative clustering

Temporal stability was evaluated over 25 consecutive weekly transitions across the simulated epidemic (Fig 5). Mean Jaccard indices across transitions were 0.876 (EDD), 0.755 (EDS), 0.851 (ESD), 0.801 (ESS), 0.888 (LD), and 0.926 (LS), indicating that the majority of cluster membership was preserved from one week to the next for all models. LS achieved the most stable partitions, with a minimum Jaccard of 0.719 across all transitions and the only model that never fell below 0.7. ESS showed the lowest minimum Jaccard (0.252), with the largest instability spike in the first epidemic week before case accumulation was sufficient to resolve cluster boundaries. ESD and LD reached minimum Jaccard values of 0.412 and 0.424, respectively. In all models, instability was concentrated in the earliest epidemic weeks when case counts and cluster structure were sparse; stability rose rapidly and remained high from week 3 onward. Mean forward and backward overlaps were consistently higher than the Jaccard index for all models, reflecting the asymmetry between new cases entering existing clusters (high backward overlap) and the survival of early cluster assignments into later partitions (variable forward overlap). Overall, all six models produced partitions that were largely stable and reproducible across the simulated epidemic horizon, supporting their use in a rolling surveillance context.

Application to empirical dataset recovers exposure-enriched clusters

Applying EpiLink (with stochastic genetic divergence) to all 297,396 unique pairs from 772 Boston sequences yielded 77 clusters of size ≥ 2, containing a total of 642 sequences (Fig 6). Cluster sizes were highly right-skewed (median = 3, mean = 8.3, SD = 14.4, max = 84). Five focus clusters contained at least one Conference- or SNF-linked case.

The largest focus cluster (cluster 11, $n = 84$) was dominated by SNF-linked cases (74 of 84 members; 88.1%), with a mean intra-cluster SNP distance of 0.69 and maximum intra-cluster distance of 4 SNPs, and a mean sampling-date gap of 2.3 days (maximum 6 days). The mean inter-cluster SNP distance was 5.0, confirming that this cluster is genetically and temporally tight relative to background clusters. The second largest focus cluster (cluster 22, $n = 29$) was enriched for Conference-linked cases (27 of 29 members; 93.1%), with mean intra-cluster SNP distance 1.14 (max 5) and mean intra-cluster temporal gap 1.4 days (max 7). Three smaller focus clusters (sizes 3, 3, and 2) also contained SNF-linked cases. Exposure composition deviated significantly from the background distribution in all five focus clusters (χ^2 goodness-of-fit, all $p < 0.003$), consistent with non-random, epidemiologically structured cluster membership. These results demonstrate EpiLink's utility for recovering known outbreak clusters with high specificity in a real-world SARS-CoV-2 surveillance dataset.

Discussion

This study introduces EpiLink, a model that quantifies the compatibility of each case pair with a user-specified latent transmission history, replacing hard threshold-based judgements with a graded measure of recent transmission plausibility. The key strength

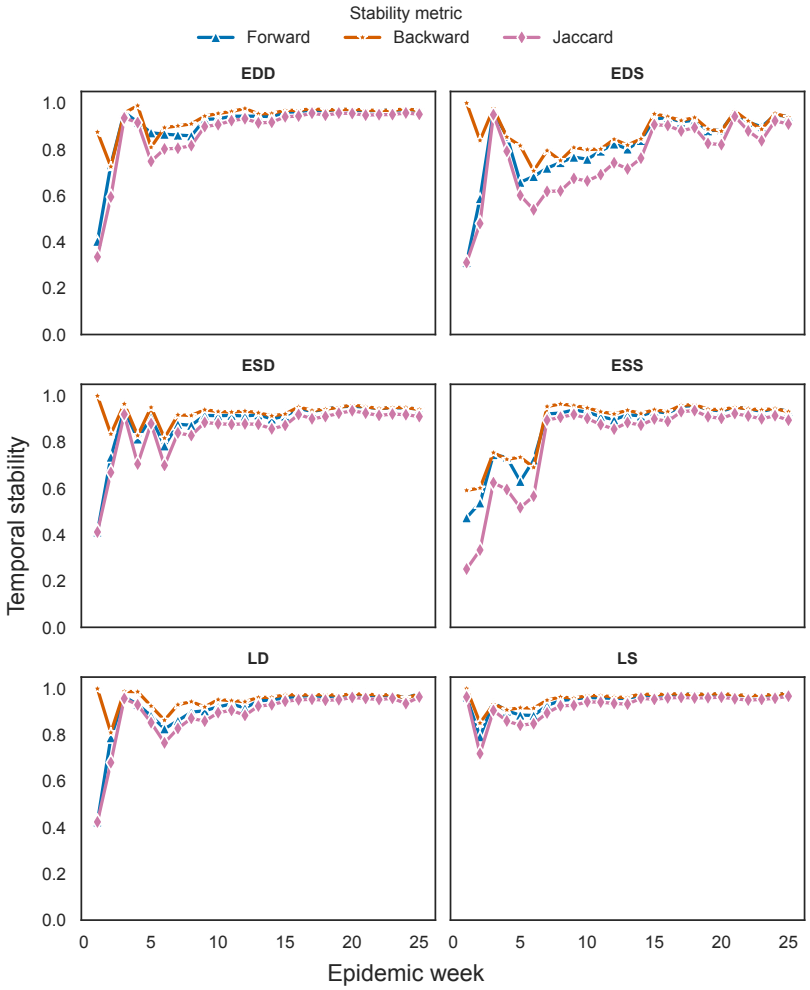

Fig 5. Temporal stability of cluster partitions as the epidemic accumulates. Forward overlap, backward overlap, and Jaccard index between consecutive weekly partitions, plotted against epidemic week. Mean Jaccard indices across all transitions: EDD = 0.876, EDS = 0.755, ESD = 0.851, ESS = 0.801, LD = 0.888, LS = 0.926. Minimum Jaccard values: EDD = 0.335, EDS = 0.311, ESD = 0.411, ESS = 0.252, LD = 0.424, LS = 0.719. Stability is lowest in the first 1–2 epidemic weeks when cluster structure is sparse, rising rapidly and remaining high thereafter. LS is the only model that never falls below Jaccard = 0.7.

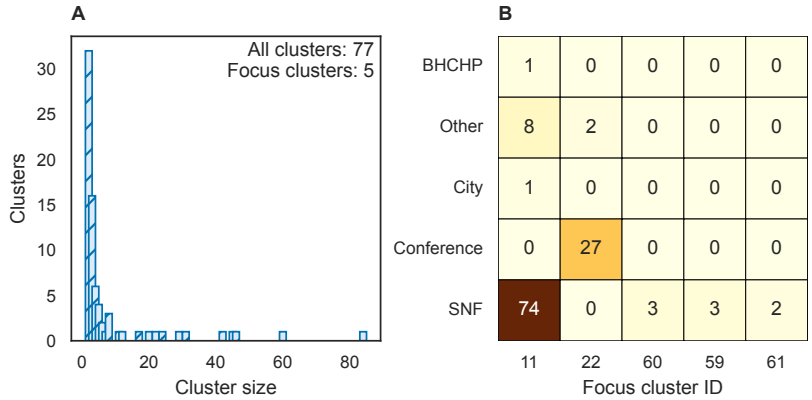

Fig 6. EpiLink clustering of the Boston MGH/DPH SARS-CoV-2 dataset. (A) Distribution of Leiden cluster sizes across all 772 sequences. Of 77 clusters (size ≥ 2), 5 are focus clusters (annotated). (B) Heatmap of exposure category counts (rows) across the 5 focus clusters (columns), annotated with raw case counts. Focus clusters contain at least one sequence linked to a Conference or skilled nursing facility (SNF) exposure. Cluster 11 ($n = 84$) is dominated by SNF cases (74/84); cluster 22 ($n = 29$) by Conference cases (27/29). Exposure composition deviates significantly from background in all five focus clusters (χ^2, all $p < 0.003$).

of this framing is interpretability: EpiLink makes explicit which latent transmission scenarios are being scored and how natural-history assumptions shape the expected distributions of temporal and genetic distances. That transparency is particularly useful in surveillance settings where analysts need to explain why a pair or cluster has been flagged.

The synthetic benchmark highlights an important limitation. Under matched baseline conditions, the supervised logistic model LD (trained on labelled pairwise data) achieved the strongest pairwise discrimination and clustering performance. This narrows the strongest defensible claim: EpiLink is best understood as an interpretable, unlabelled, mechanistic scorer of recent-transmission compatibility, not a universally superior predictive classifier. Within that role, the heterogeneity across EpiLink variants is informative in its own right: EDD, EDS, ESD, and ESS each capture distinct forms of temporal-genetic model mismatch rather than being interchangeable implementations of the same score.

Performance degradation under misspecification partly reflects inherent limits of the genomic data rather than model-specific failures. Incomplete sampling constrains how much transmission structure can be resolved regardless of the method used [41]. When incubation period variability is underestimated or substitution rates are low, near-identical consensus genomes can arise from direct transmission, unsampled chains, or co-primary infection alike, and temporal information can only partially resolve this ambiguity [18, 34, 42–44]. The sensitivity results therefore highlight which epidemiological and evolutionary conditions most limit pairwise identifiability, regardless of the scoring approach.

Graph sparsification provides an important practical result. A low compatibility threshold retained nearly all total edge weight while discarding the large majority of edges, yielding substantial runtime reductions with no meaningful effect on downstream community structure. This threshold emerged consistently across all model variants and the full synthetic dataset, suggesting it reflects the intrinsic concentration of compatibility weight in a small fraction of pairs rather than a model-specific artefact, and opens the way for routine application to large genomic surveillance datasets.

The temporal stability and Boston analyses clarify where EpiLink is most compelling. Once cumulative surveillance extended beyond the earliest sparse weeks, cluster assignments were largely preserved from one weekly update to the next. In the Boston application, stochastic EpiLink recovered clusters strongly enriched for the skilled nursing facility and conference outbreaks without any supervised training. This is the operational context in which EpiLink is most useful: consensus genomes and case times are available, exact transmission labels are not, and the goal is to recover interpretable recent-transmission neighbourhoods rather than direction-resolved transmission edges.

Three design choices deserve mention. The synthetic benchmark assessed performance both with and without corresponding updates to inference parameters, separating best-case (matched) from realistic (misspecified) conditions. The positive-pair definition and clustering evaluation were designed to capture local recent-transmission neighbourhoods rather than complete outbreak labels. And the default target was deliberately restricted to direct transmission and co-primary infection; expanding it to include unsampled intermediates would broaden the score but reduce specificity for close linkage. These are principled choices, not incidental implementation details.

Taken together, these results point to a practical middle ground in genomic epidemiology. When labelled training data are abundant and transferable, supervised models may offer stronger pairwise discrimination. EpiLink remains valuable because it provides an interpretable alternative to ad hoc distance thresholds, makes its biological assumptions explicit, and can recover meaningful recent-transmission structure from the joint information in case timing and consensus genomes—especially when the inferential target is recent linkage rather than maximal clustering accuracy.

Supporting information

S1 Appendix. Detailed derivation and modelling assumptions for the EpiLink compatibility model. Provides the full derivation of the scenario-specific testing-time difference and transmission-related branch length used by EpiLink, together with the natural-history, observation, and molecular-clock assumptions.

S1 Fig. Sensitivity of average precision (AP) to perturbations of data-generating parameters. Lollipop plot showing AP loss relative to the unperturbed baseline for each model across twelve perturbation scenarios, under matched (blue) and mismatched (coral) inference conditions. Matched: inference parameters updated to reflect each perturbed scenario; mismatched: inference parameters held fixed at baseline. Each row corresponds to a perturbation scenario; the multiplier applied to the indicated parameter is shown on the left. Positive values indicate improved performance relative to baseline; negative values indicate degradation. Arrows indicate values exceeding the axis range. Model codes follow Table 2; LD and LS were trained on labelled transmission pairs and serve as a supervised reference.

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Competing interests

The authors have declared that no competing interests exist.

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